Implementation of fish farming and documentation of knowledge possessed by members of the working group regardin dish species and their culinary art.

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Summary

One of the main problems suffered by Lake Pátzcuaro is overfishing, the loss of aquatic diversity and the deterioration of ancestral knowledge and knowledge. In the present study, we will work on the implementation of a fish farming system of native species of the lake, documenting each of the production stages. The objective is to integrate the knowledge and wisdom of the community of Santa Fe de la Laguna, which will help recover local knowledge. This will benefit the community by providing access to food and will allow for a better understanding of the biological knowledge of the species throughout the production process.

Keywords: fish farming, knowledge, native species, production systems.

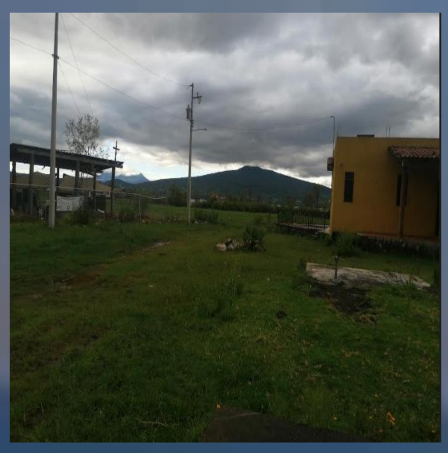
Introduction

During the last decades and currently (2024), serious world order problems have been recorded, such as deforestation, the greenhouse effect, desertification and loss of biodiversity. Therefore, it is necessary to promote better conservation and sustainable use of biodiversity, by promoting adequate use of the water resource, adapting to climate change, stopping the increase in the level of nutrients in water and soils, as well as the valuation of the contributions of biodiversity to human well-being. In this last area, it places indigenous communities as the key actors, since in addition to being the most vulnerable sector to ecological degradation, they have a set of knowledge and use practices about the environment that is based on the principles of sustainability, product of a long history of interaction and knowledge of their natural environment (Puc Gil and Retana Guiascón, 2012) and conceptualization of their well-being based on good living "sesi irekani" in synchrony with nature, and in contrast to capitalist development (Alvarado Pizaña, 2020).

Therefore, the objective is to investigate the traditional knowledge of fishermen and people from the community of Santa Fe de la Laguna about the endemic fish of the lake, so that they are integrated with scientific knowledge about aquaculture; with the purpose of developing a proposal for participatory community fish farming that contributes to the conservation and strengthening of community nutrition, by having this source of food production.

Preliminary Results

Stage 1. Adaptation of the production system



Methods

In this project, a semi-intensive fish production system will be implemented, for which a recirculating aquaculture system (RAS) has been planted, aimed at the design and development of a participatory community aquaculture strategy, in which traditional knowledge will be considered, as well as the different stages or productive processes of the acúmara Algansea lacustris and kurucha urapiti Chirostoma estor.

STAGE1. COMMUNITY PARTICIPATION

- Initial and permanent training.
- Tank conditioning.

Stage 2. Cultivation of native species

- Fish transports
- tocking of fish
- Logs:
  - Water quality record.
  - Mortality registry
  - Registration of biological parameters (length and weight).
- Food: adjustment to satiety.

Stage 3. Harvest and marketing.

- Harvest.
- Fish sale.
- Disinfection of the tanks.

Preliminary Conclusions

Local knowledge is of utmost importance since it is oriented towards the care of the environment and the sustainable use of natural resources.

At the same time, it should be noted that the knowledge that Fishermen possess is of great help for the management of a fish production system since they know a lot about the species.

At the same time, we recognize the participation in this Project of the indigenous women who are the key actor of this study and that much depends on them since for the final stage they are the ones in charge of preparing the ancestral foods and it is very necessary to continue to improve the health and nutrition of the community.

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